HARDWARE × MODELS × REAL-WORLD SPEED

Intelligence. On your desk.

A field guide to capable local coding and tool-using agents.
14 hardware configurations. Seven local models + four Claude Opus reference points.

Start with the model.
Then choose memory, speed and budget.
Practical local-agent setups, from a laptop to a deskside supercomputer.

THE PORTABLE PICK
M5 Max · 128 GB
Flash-Next + MTPLX: **64 tok/s reported** at 65K context. Its n-gram table streams from SSD.

THE CAPABLE DESK-SIDE PICK
M5 Ultra · 256 GB
GLM-5.3-Flash at 44–62 tok/s with DFlash2 in oMLX. The 512 GB model adds GLM-5.3.

THE COMPACT CUDA PICK
DGX Spark · 128 GB
Flash-Next at **37–46 tok/s reported**. Two Sparks run higher-scoring GLM-5.3-Flash at **28–63** with DFlash2.

01 / The hardware-to-model map

M reported measurement · E estimate · Bottleneck: memory · compute · other

35 tok/s gate: choose the highest-scoring fitting model whose decode range reaches ≥35 tok/s, then sort by score and speed ↑. One active request; a qualifying upper bound is not a guaranteed minimum.

DEVICE / CONFIGURATION	APPROX. USD	MEMORY INSTALLED	BANDWIDTH GB/S	AI COMPUTE TFLOPS / TOPS	BEST-FIT SHORTLIST MODEL / PRECISION	DECODE TOKENS / SEC	MEMORY TIME	COMPUTE TIME	OTHER TIME
MacBook Pro · M5 [H1] 10-core GPU	**$2.5k** System	**32 GB** Unified	153	**~16 TFLOPS** E FP16 GPU · INT8 ~30	Qwen3.6-35B-A3B · 4-bit MLX oMLX 0.3.6 · ~20 GB peak	**51–60** M 1K–4K · two runs · B12	**59–67%**	20–23%	33–41%
MacBook Pro · M5 Pro [H1] 20-core GPU	**$3.7k** System	**64 GB** Unified	307	**~36 TFLOPS** E FP16 GPU · INT8 ~67	Qwen3.8-27B · dynamic Q4 MTPLX · native MTP drafting	**21–44** E Upper end: repetitive editing	**45–94%**	**23–48%**	6–55%
GeForce RTX 4090 [H3]	**$2–2.5k** Card est.	**24 GB**	1,008	**661 TFLOPS** FP8	Qwen3.8-27B · Q4_K_S llama.cpp · no MTP	**38–46** M 64K → 4K · B10	**83–85%**	7%	15–17%
Radeon AI PRO R9700 [H4]	**$1.7k** Card	**32 GB**	640	**766 TOPS** INT4 · 383 TFLOPS FP8	Qwen3.8-27B · Q4-class + MTP llama.cpp Vulkan · MTP=2	**51.8** M AMD vendor · preliminary · B8	**58%**	10%	42%
GeForce RTX 5090 [H5]	**$6.4–7.4k** Card est.	**32 GB**	1,792	**1,676 TFLOPS** FP4	Qwen3.8-27B · Q4_K_M LM Studio / CUDA	**59** M 6.1K · no MTP · B3	**59%**	7%	41%
Ryzen AI Max+ 395 / Halo [H2] Radeon 8060S	**$2–4k** System est.	**128 GB** GPU share varies	256 LPDDR5X-8000	**50 TOPS** NPU · INT8	Qwen3.8-Flash-Next · UD-IQ4_XS llama.cpp ROCm · MTP + n-gram drafting	**32–47** M 8K new code → rewrite · B7	32–34%	8–9%	**66–68%**
1 × DGX Spark [H7] Single NVIDIA GB10 system	**$4.7k** System	**128 GB**	273	**500 TFLOPS** FP4	Qwen3.8-Flash-Next · NVFP4 vLLM + MTP3 · table on SSD	**37–46** M Prose → code · B5	35–44%	4–5%	**56–65%**
MacBook Pro · M5 Max [H1] 40-core GPU	**$6k** System est.	**128 GB** Unified	614	**~70 TFLOPS** E FP16 GPU · INT8 ~130	Qwen3.8-Flash-Next · dynamic Q4 MTPLX 2.12 + MTP · n-gram table on SSD	**63.5** M 65.5K prompt · B1	28%	13%	**72%**
RTX PRO 6000 Blackwell [H6] Workstation Edition	**$16–20k** Card	**96 GB**	1,792	**2,015 TFLOPS** FP4	Qwen3.8-Flash-Next · NVFP4 vLLM · n-gram table in ~100 GB host RAM	**81–88** M No MTP; 120–173 MTP · B9	15–17%	2%	**83–85%**
2 × DGX Spark [H7]	**$9.4k** Systems	**256 GB** 128 GB per node	2 × 273 Distributed	**1,000 TFLOPS** FP4 · summed	GLM-5.3-Flash · NVFP4 / EXL3 4-bit TP2 · DFlash2 drafter · 200G RoCE	**28–63** M Prose → code · B16	**64–66%**	10%	34–36%
Mac Studio · M5 Ultra [H8] 80-core GPU	**$12k** System est.	**256 GB**	1,200	**~140 TFLOPS** E FP16 GPU · INT8 ~260	GLM-5.3-Flash · oQ4e / mixed 4/8-bit oMLX + DFlash2 drafter	**44–62** M 4K → short prompt · B17	28–35%	14–18%	**65–72%**
2 × RTX PRO 6000 Blackwell [H6]	**$32–40k** Cards est.	**192 GB** 96 GB per GPU	2 × 1,792 Not shared	**4,030 TFLOPS** FP4 · summed	GLM-5.3-Flash · IQ4_XS Layer split · ~157 GB weights · fit projected	**41–70** E Best ~55 · B13 anchor	25–42%	3–5%	**58–75%**
M5 Ultra · 512 GB [H8] Ships late October	**$15–18k** System est.	**512 GB**	1,200	**~140 TFLOPS** E FP16 GPU · INT8 ~260	GLM-5.3 · IQ4_XS / oQ4 Needs DFlash2 drafter · ~365–420 GB weights	**20–40** E Drafting · 15–25 without	27–55%	15–30%	**45–73%**
DGX Station · GB300 [H9] OEM order · deskside	**$99–124k** System	**748 GB** 252 HBM + 496	7,100 / 396 HBM / LPDDR5X	**15,000 TFLOPS** FP4 vendor	GLM-5.3 · IQ4_XS, split HBM / CPU Needs DFlash2 drafter + split-memory backend	**21–40** E Drafting · 16–25 without	**38–72%**	7–13%	28–62%

Prices: approximate USD before tax; cards exclude the host PC; "est." means a budget estimate. **Speeds:** externally reported (M) or estimated (E); model, context and runtime differ. **Compute:** dense matrix peaks of the GPU, or the NPU where labeled. Apple has no published figure, so its E values are third-party estimates for the matrix units built into each M5 GPU core. Peak compute mostly drives prefill; single-user decode is set by bandwidth, which is why the Macs decode well despite lower peaks. **Memory / Compute / Other time:** share of each generated token's measured time that its memory reads alone would take at peak bandwidth, that its arithmetic alone would take at peak vector compute, and that neither explains (launches, synchronization, host lookups, slow kernels). Memory and compute overlap, so the three do not add up. The shaded cell is the bottleneck: the largest share, with Memory and Compute both shaded when compute is at least half of memory. Shaded Memory means faster memory helps; Compute, faster arithmetic; Other, better software. Derived from a roofline in the drawer.

02 / Local models & the frontier trajectory

BASELINE Claude Opus · cloud yardstick, not runnable locally — AA Intelligence Index v4.3.2 · higher is better

10 independent evaluations: agents 30% · coding 20% · scientific reasoning 20% · general 30%. **Opus trajectory:** 4.6 (Feb) 32* → 4.8 (May) 42 → 5 (Jul) 51 → 5.5 (Sep) 58. The best local-fitting model, GLM-5.3 (45), sits between May's Opus 4.8 and July's Opus 5.

MODEL / TESTED SETTING	AA SCORE	PARAMETERS / TYPE	MEMORY ESTIMATE · 128K CONTEXT · 1 ACTIVE REQUEST
Claude Opus 5.5 BASELINE Max · Sep 2026 · cloud ceiling	58	**Undisclosed** Architecture undisclosed	**Cloud only · weights and serving memory undisclosed**
Claude Opus 5 BASELINE Max · Jul 2026 · cloud reference	51	**Undisclosed** Architecture undisclosed	**Cloud only · frontier trajectory reference**
GLM-5.3 Max	45	**~753B · MoE** ~40B active	**365 GB weights + ~13 GB cache ≈ 378 GB resident** IQ4_XS · compressed MLA, 16-bit cache · SSD tables: none · runtime: +8–16 GB
Claude Opus 4.8 BASELINE Max · May 2026 · cloud reference	42	**Undisclosed** Architecture undisclosed	**Cloud only · frontier trajectory reference**
GLM-5.3-Flash Reasoning	42	**320B · MoE** 18B active	**157–200 GB weights + ~2 GB cache ≈ 159–202 GB resident** Q4-class · compressed MLA, 16-bit cache · SSD tables: none · runtime: +8–16 GB Expanded-KV backends need ~95–105 GB cache instead.
Qwen3.8-Flash-Next Reasoning	40	**~180B · MoE + lookup** 125B backbone · 6B active	**~83 GB weights + ~4 GB cache ≈ 87 GB resident** MTPLX dynamic Q4 · 16-bit cache · +32 GB lookup tables on SSD · runtime: +4–8 GB
DeepSeek V4.1 Flash Max	39	**~763B · MoE + Engram** 552B backbone · 16B active decode	**~316 GB weights + ~0.2–0.6 GB cache ≈ 317 GB resident** DwarfStar Q4 · packed FP4–BF16 cache scenarios · +203 GB Engram on SSD · runtime: +8–16 GB
Qwen3.8-27B Xhigh	34	**27B · Dense** 27B active	**20.4 GB weights + ~9.3 GB cache ≈ 30 GB resident** MTPLX dynamic Q4 · 16-bit cache · SSD tables: none · runtime: +4–8 GB
DeepSeek V4 Flash / 0731 Max · text model	34	**284B · MoE** 13B active	**~155 GB weights + ~1 GB cache ≈ 156 GB resident** Q4_K_XL · compressed attention, BF16 cache · SSD tables: none · runtime: +8–16 GB
Claude Opus 4.6 BASELINE Max · Feb 2026 · AA estimate	32	**Undisclosed** Architecture undisclosed	**Cloud only · frontier trajectory reference** AA marks 32 as estimated; independent evaluation forthcoming.
Qwen3.6-35B-A3B Reasoning · fallback	18	**35B · MoE** ~3B active	**~22 GB weights + ~3 GB cache ≈ 25 GB resident** Q4 · 16-bit cache · runtime: +4–8 GB